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RESEARCH ARTICLE

A Multi-Camera Temporal Fusion for False Alarm Suppression in Edge-Based Weapon Surveillance

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ABSTRACT Real-time weapon detection is a critical component of intelligent surveillance systems, particularly for perimeter monitoring applications on embedded edge platforms. However, reliable alarm generation remains challenging because false positives, temporal instability, and viewpoint inconsistencies can propagate through conventional multi-camera fusion strategies. To address these limitations, this work proposes a lightweight Adaptive Multi-Camera Temporal Fusion (ACTF) framework that combines confidence-aware evidence separation, temporal persistence, and short-window cross-camera validation at the decision level, thereby confirming detections without requiring additional neural-network inference. The framework was evaluated using a TensorRT-optimized YOLO26s detector in controlled dual-camera scenarios involving clear visibility, partial occlusion, visually ambiguous distractors, and challenging illumination. While logical OR fusion achieved higher recall, it also propagated erroneous detections; in contrast, ACTF completely suppressed the distractor-induced false alarms while maintaining competitive performance and sub-second confirmation under favorable conditions. The original NVIDIA Jetson Nano implementation achieved an average throughput of 2.4 camera-pair cycles per second, corresponding to low-rate online embedded operation, whereas an additional NVIDIA Jetson Xavier NX benchmark achieved an average of 10.2 camera-pair cycles per second. This is equivalent to 10.2 processed frames per second for each camera stream and 20.4 camera images per second in aggregate. These results support reactive real-time embedded operation on the Xavier NX platform and demonstrate that ACTF improves alarm reliability with negligible decision-level computational overhead.

INDEX TERMS Weapon detection, multi-camera, edge AI, embedded systems, reactive real-time processing, latency-constrained systems, YOLO.

I. INTRODUCTION

In recent years, deep learning has significantly advanced the field of computer vision, particularly in object detection tasks. Modern deep neural networks enable accurate detection and localization of objects in both images and video streams, supporting a wide range of real-world applications such as intelligent surveillance, smart cities, and public safety monitoring [1], [2], [3]. These advances are primarily driven by convolutional neural networks (CNNs), which are capable of learning hierarchical feature representations and consistently outperform traditional computer vision approaches.

Object detection has become a core component of intelligent surveillance systems, where automated analysis of visual data is required to identify critical events in real

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time. In this context, modern detection frameworks are typically categorized into two-stage and one-stage detectors. While two-stage detectors achieve high accuracy through region proposal mechanisms, one-stage detectors such as the YOLO family and SSD provide a more favorable balance between accuracy and computational efficiency, making them particularly suitable for real-time applications [4], [5]. The YOLO family, in particular, has evolved significantly across multiple versions, achieving strong performance in terms of both detection accuracy and inference speed [6].

Among surveillance-related applications, real-time weapon detection has emerged as a critical research area due to its direct impact on public safety and threat prevention. Automated systems aim to identify firearms and sharp objects in dynamic environments, enabling early warning mechanisms in security-critical scenarios. However, detecting such objects in real-world conditions remains challenging due to factors such as occlusions, viewpoint variations, scale changes, and illumination variability [7], [8], [9]. Recent systematic reviews have highlighted that, despite significant advances in deep learning-based detection, false positives and unreliable detections remain persistent challenges in real-world surveillance scenarios [10]. In addition, the high cost of false-positive alarms requires detection systems to achieve not only high accuracy but also high reliability.

Recent works have explored the integration of deep learning and edge computing to enable real-time detection directly on embedded devices, reducing dependency on cloud-based processing while improving privacy and response time [11], [12], [13]. These studies demonstrate the feasibility of deploying intelligent surveillance systems in practical environments. However, they still present important limitations in terms of detection latency and system responsiveness, with reported processing times often exceeding one second per detection. Such delays can hinder timely decision-making in critical scenarios and highlight the need for more efficient solutions tailored to edge-based real-time applications.

To address these limitations, optimized inference frameworks such as TensorRT have been shown to significantly improve latency and hardware efficiency for YOLO-based detectors on NVIDIA Jetson platforms [14]. Similarly, edge-based smart surveillance systems have demonstrated the capability of running artificial intelligence algorithms on low-power embedded devices, even in distributed camera configurations [15]. Nevertheless, maintaining robust detection performance under multi-camera configurations remains challenging due to increased computational load and the need for efficient fusion strategies.

Despite these advances, most existing approaches rely on single-camera configurations, which are inherently limited by restricted fields of view and susceptibility to occlusions. Multi-camera systems provide complementary viewpoints and improved spatial coverage, enhancing detection robustness by reducing blind spots and mitigating occlusions across perspectives [16]. Prior research in multi-camera vision and tracking has demonstrated the benefits of integrating information from multiple viewpoints, while also highlighting challenges such as appearance variability, synchronization, and multi-camera data association [17], [18]. However, integrating multiple camera streams while preserving low-latency and computationally efficient operation on resource-constrained edge platforms remains an open challenge.

Simple multi-camera decision rules do not provide a robust balance between detection sensitivity and false-alarm suppression. This limitation motivates the development of a lightweight temporal validation strategy capable of improving alarm reliability without modifying the underlying detector architecture.

Unlike prior works that primarily focus on improving detector-level accuracy, this study shifts the focus toward system-level reliability in real-world surveillance deployments. In particular, this work addresses the problem of false alarm propagation in multi-camera embedded systems by introducing a lightweight validation mechanism that operates at the decision level.

To this end, this work proposes a lightweight Adaptive Multi-Camera Temporal Fusion (ACTF) strategy for false-positive suppression and robust alarm confirmation in embedded dual-camera weapon surveillance scenarios. The proposed method integrates three complementary mechanisms: confidence-aware evidence separation, temporal persistence, and short-window multi-camera validation. Rather than maximizing raw frame-level detection sensitivity, the objective is to improve alarm trustworthiness by filtering transient and ambiguous detections before issuing a final confirmed alarm.

The scenario-based evaluation was implemented using a TensorRT-optimized YOLO26s detector on an NVIDIA Jetson Nano with dual-camera inputs, while an additional computational benchmark was conducted on an NVIDIA Jetson Xavier NX using the same detector configuration. In contrast to computationally intensive multi-view deep learning approaches, ACTF operates as a lightweight decision-level validation layer and does not require an additional neural-network inference stage. Consequently, its computational overhead is negligible relative to detector inference and camera-processing operations.

The Jetson Nano implementation achieved an average of 2.4 camera-pair cycles per second and is therefore characterized as low-rate online embedded operation. In contrast, the Jetson Xavier NX implementation achieved an average of 10.2 camera-pair cycles per second, equivalent to 10.2 processed frames per second for each camera stream and 20.4 camera images per second in aggregate. This performance supports reactive real-time embedded operation while remaining distinct from conventional 25–30 FPS full-frame-rate surveillance processing. The main contributions of this work are summarized as follows:

- A lightweight dual-camera temporal alarm confirmation framework specifically designed for false-positive suppression in embedded weapon surveillance systems.

- A decision-level multi-camera verification strategy integrating temporal persistence and multi-camera confirmation without modifying the underlying detection architecture.
- A scenario-based evaluation demonstrating complete false-alarm suppression under distractor-based conditions, together with a cross-platform computational analysis showing low-rate online operation on the Jetson Nano and reactive real-time processing on the Jetson Xavier.

II. RELATED WORK

A. DEEP LEARNING AND EDGE-BASED WEAPON DETECTION

In recent years, deep learning–based approaches have significantly advanced automatic object detection in surveillance environments. Several review studies have highlighted the effectiveness of convolutional neural networks and YOLO-based architectures for detecting objects in complex scenes, achieving a favorable balance between detection accuracy and inference speed. Recent surveys indicate that deep learning methods generally outperform conventional machine learning approaches, particularly in dynamic and cluttered environments [2], [19]. Reviews focused specifically on surveillance footage have identified YOLO and Faster R-CNN as two of the most widely adopted detector families, although small objects, occlusion, illumination variability, and visually similar distractors remain important challenges [10].

Alternative sensing modalities have also been investigated for weapon detection. Veranyurt and Sakar proposed a deep learning approach for concealed-pistol detection using thermal imagery, demonstrating the potential of non-visible spectrum data under challenging observation conditions [20]. Although alternative sensors may improve object visibility, these approaches primarily address detector-level performance and do not directly resolve alarm-level problems such as temporal instability, false-positive propagation, or cross-camera inconsistency.

Several studies have explored the deployment of object-detection models on resource-constrained edge platforms. Berardini et al. implemented a deep learning framework on an NVIDIA Jetson Nano for detecting small objects in indoor surveillance scenarios [21]. Lightweight network optimization has also been investigated to reduce model complexity. Sun et al. introduced YOLO-E for military target recognition, obtaining a favorable complexity–accuracy trade-off [22]. Similarly, super-resolution processing combined with YOLO-based detection has been used to improve small-object recognition on embedded devices [23].

Other approaches employ hybrid or multi-stage pipelines to improve detection robustness. Deep convolutional networks combined with classical preprocessing have been proposed to improve detection under challenging conditions [24]. However, additional preprocessing, enhancement, or neural-network stages may increase latency, memory consumption, and implementation complexity.

These studies primarily improve the detector architecture, input representation, or feature-extraction process. Consequently, their performance gains do not necessarily prevent an isolated high-confidence misclassification from being propagated as a system alarm. This distinction is particularly relevant in embedded surveillance, where alarm reliability must be improved without adding substantial neural-network complexity.

B. MULTI-CAMERA SURVEILLANCE AND DECISION FUSION

Multi-camera surveillance has received increasing attention because complementary viewpoints can improve scene coverage and provide additional evidence when one camera is affected by occlusion or an unfavorable perspective. Multimodal and spatio-temporal fusion techniques have demonstrated improved performance in complex environments [25]. Cross-camera tracking and feature-fusion methods have also shown that maintaining identity, spatial, and temporal consistency across viewpoints remains challenging because of appearance variations, perspective transformations, and different observation conditions [26].

Logical AND and OR rules remain common baseline strategies because of their low computational cost. OR fusion activates a positive response when either camera reports a detection, generally improving sensitivity but also propagating errors produced by a single viewpoint. AND fusion requires simultaneous positive responses and may therefore suppress valid detections when one viewpoint is affected by occlusion, illumination, or perspective limitations [17].

These limitations indicate that multi-camera availability alone does not guarantee reliable alarm generation. A practical fusion strategy must determine whether detections are persistent, temporally compatible, and supported by complementary viewpoints, while preserving the low computational cost required for embedded deployment.

C. TEMPORAL FALSE-ALARM REDUCTION STRATEGIES

Temporal post-processing has been widely investigated as a mechanism for suppressing transient false detections. Kong et al. computed a moving average of frame-level fire probabilities over a fixed temporal interval. Their experiments showed that temporal smoothing reduced false alarms, although larger smoothing intervals introduced a reduction in sensitivity and an increase in detection delay [27].

Recursive filtering provides an alternative to fixed-window averaging. Wang et al. employed an online Kalman filter to smooth reduced-reference image features and their temporal derivatives for camera-anomaly detection. The Kalman-filtered implementation reduced false positives from seven to two and increased alarm precision from 88.89% to 96.55%, while maintaining a recall of 98.25%. Their

experiments also indicated that the filtering stage introduced negligible computational overhead [28].

Hysteresis thresholding has also been established as a structured surveillance-processing strategy. Yeh et al. proposed a three-pronged framework combining color and texture background modeling, spatial compensation through hysteresis thresholding, and temporal compensation using motion history. Their method generates a predominant foreground representation using a restrictive threshold and supplementary representations using progressively less restrictive thresholds. Supplementary evidence is incorporated only when spatially connected to the predominant foreground, preventing unrestricted acceptance of noisy low-threshold responses [29].

Yeh et al. evaluated their method against GMM, MRF, LBP, ViBE, PBAS, CP3, and MST on several established surveillance datasets. The proposed combination obtained competitive or improved precision, recall, similarity, and F-measure in several scenarios. For example, it achieved a recall of 0.970, precision of 0.901, and F-measure of 0.933 for an object moving toward the camera. These results demonstrate that separating restrictive and permissive evidence through hysteresis is an established surveillance strategy rather than an ad hoc thresholding rule [29].

Gao et al. proposed another spatio-temporal post-processing framework for aerial vehicle detection. Their spatial stage combines multi-neighborhood hysteresis thresholding with morphological opening and closing, whereas the temporal stage evaluates object consistency between adjacent registered frames using intersection-over-union and centroid-distance criteria. Their experiments reported an average false-positive reduction of 83.8% across the evaluated detection algorithms, although false-positive suppression was accompanied by some loss of true positives [30].

Sliding-window voting has also been applied to alarm validation. Zaidi et al. analyzed safe and unsafe classifications over a fixed 5-s interval and generated an alarm when at least 70% of the frames were classified as unsafe. Their temporal-analysis module increased the average monitoring accuracy from 97.97% to 100% across five evaluated videos by preventing isolated missed or false detections from directly generating alarms [31].

These approaches demonstrate that temporal filtering can be implemented through confidence averaging, recursive state estimation, hysteresis, inter-frame spatial consistency, or window-based voting. However, they primarily regularize evidence within one stream, between adjacent frames, or at the pixel, feature, object-shape, or scene-label levels. They do not explicitly require class-consistent strong evidence from two complementary camera viewpoints before confirming an alarm.

Table 1 summarizes the main temporal mechanisms, evidence sources, reported findings, and methodological differences between representative false-alarm reduction strategies and the proposed ACTF method. The published numerical values are included to characterize each original study and are not interpreted as a direct performance ranking because the methods were evaluated on different tasks, datasets, and hardware platforms.

As summarized in Table 1, temporal smoothing, recursive filtering, hysteresis, motion-history compensation, inter-frame spatial consistency, and sliding-window voting are established strategies for improving surveillance reliability. In particular, the results of Yeh et al. demonstrate that the use of restrictive and supplementary thresholds is a formally validated surveillance mechanism rather than an ad hoc thresholding decision [29].

ACTF differs primarily in the source and interpretation of the evidence. The reviewed approaches regularize a probability, feature trajectory, foreground mask, object correspondence, or scene-level classification within one stream or between adjacent frames. ACTF instead requires class-consistent strong evidence from two complementary viewpoints within a timestamp-based temporal interval before confirming the alarm.

D. RESEARCH GAP AND POSITIONING OF ACTF

The reviewed literature confirms that temporal smoothing, recursive filtering, and hysteresis are established mechanisms for improving surveillance reliability. Nevertheless, the source and interpretation of temporal evidence differ from those used by ACTF. Moving-average and Kalman-based methods smooth a single temporal signal. Pixel-level hysteresis and morphological methods improve foreground segmentation, while sliding-window approaches aggregate scene labels over a fixed interval.

ACTF adopts the established principle of separating reliable and supplementary evidence but transfers it to a confidence-aware, decision-level, multi-camera formulation. The weak threshold T_w allows uncertain detections to contribute to a pre-alert, whereas the strong threshold T_s identifies detections eligible for cross-camera confirmation. Temporal persistence prevents isolated weak responses from generating alarms, and timestamp-based validation verifies whether class-consistent strong evidence from the complementary camera occurred within Δt.

Therefore, ACTF should not be interpreted as a conventional pixel-level hysteresis algorithm, a confidence-smoothing filter, or an object tracker. It is a lightweight alarm-validation layer that combines the general hysteresis principle with temporal persistence and multi-camera confirmation. Unlike complex multi-view neural architectures, it does not modify the underlying YOLO26s detector or require an additional inference stage.

The identified research gap is thus the limited availability of computationally lightweight mechanisms that jointly integrate confidence separation, temporal state accumulation, class consistency, and cross-camera timestamp validation for embedded weapon-surveillance alarms. The proposed ACTF strategy addresses this gap while explicitly recognizing that strict complementary-view confirmation may reduce recall

TABLE 1. Comparison of representative temporal and hysteresis-based strategies for false-alarm reduction in video surveillance.

Approach	Temporal or decision mechanism	Evidence employed	Cross-camera confirmation	Reported finding	Relationship with ACTF
Moving-average temporal smoothing [27]	Average of classification probabilities over a fixed K-frame interval	Frame-level fire probability from one surveillance stream	No	Temporal smoothing reduced false alarms. With $K = 30$, the method reported a 98.5% fire-detection rate, while larger smoothing intervals reduced false alarms at the expense of sensitivity and response delay.	Both methods reject transient observations. However, moving-average filtering modifies a single confidence signal, whereas ACTF preserves weak and strong evidence separately and additionally requires cross-camera validation.
Online Kalman filtering [28]	Recursive state prediction and measurement correction in the temporal domain	Image-quality features and their temporal derivatives	No	The Kalman-filtered implementation reduced false positives from 7 to 2 and increased precision from 88.89% to 96.55%, while maintaining a recall of 98.25% and a processing rate of 20.8 FPS.	Kalman filtering suppresses feature and measurement noise, but a semantically incorrect detection may still form a smooth temporal trajectory. ACTF instead verifies class-consistent evidence from two camera viewpoints.
Three-pronged hysteresis and motion compensation [29]	Predominant and supplementary thresholds combined with spatial shape recovery and motion-history compensation	Color, texture, foreground connectivity, and detections from the two preceding frames	No	The method achieved competitive or superior precision, recall, and F-measure over GMM, MRF, LBP, ViBE, PBAS, CP3, and MST in several surveillance sequences. For an object moving toward the camera, it reported recall 0.970, precision 0.901, and F-measure 0.933.	This work demonstrates that hysteresis is an established surveillance strategy rather than an improvised threshold rule. Its hysteresis acts at pixel and foreground-shape levels, whereas ACTF transfers the strong/weak-evidence principle to multi-camera alarm validation.
Spatio-temporal aerial processing [30]	Multi-neighborhood hysteresis, morphological filtering, and adjacent-frame object consistency	Foreground intensity, morphology, intersection-over-union, and centroid displacement	No	The complete spatio-temporal scheme produced an average reduction of 83.8% in false positives and improved the detection F-score across multiple aerial-vehicle detection algorithms, although some true positives were also removed.	The method requires registered consecutive frames and geometric correspondence between objects. ACTF operates at the decision level and does not require image registration, bounding-box overlap, or centroid association.
Sliding-window temporal voting [31]	Alarm activation when a predefined percentage of frames within a fixed time window are classified as unsafe	Safe/unsafe scene classifications from one monitoring stream	No	Using a 5-s window and a 70% unsafe-frame threshold increased average monitoring accuracy from 97.97% to 100% across five videos, preventing isolated classification errors from generating alarms.	Window voting is suitable for conditions that persist for several seconds but may delay short or rapidly developing events. ACTF uses recursive updates and a shorter timestamp interval of $\Delta t = 0.6$ s.
Proposed ACTF	Weak/strong confidence separation, pre-alert persistence, timestamp-based compatibility, and confirmation accumulation	Predicted class, confidence, and timestamps from two complementary camera viewpoints	Yes	ACTF suppressed the 12 false-positive responses propagated by logical OR fusion in the controlled distractor sequence, producing $FP = 0$ and specificity $= 1.000$. This improvement was accompanied by lower recall under partial occlusion and challenging illumination.	ACTF extends the established temporal and hysteresis principles to a lightweight decision-level multi-camera mechanism. Its principal limitation is that a valid single-view detection may remain unconfirmed when the complementary viewpoint is degraded.

when one camera is degraded by occlusion, illumination, or perspective.

III. METHODOLOGY

A. DATASET

1) DATASET DESCRIPTION

The experimental evaluation was conducted using the Sohas Weapon Detection Dataset, obtained from the OD Weapon-Detection repository [32]. This dataset contains annotated images depicting firearms and bladed objects under diverse conditions representative of real-world video surveillance scenarios, including variations in illumination, background complexity, object scale, and partial occlusions.

The dataset comprises a total of 5,859 annotated images, partitioned into training, validation, and test subsets. Specifically, 4,252 images were used for training, 750 images for validation, and 857 images for testing. Each image is associated with a corresponding annotation file in YOLO format, ensuring a one-to-one correspondence between images and labels across all subsets and enabling seamless integration with modern YOLO training pipelines.

Table 2 presents the distribution of instances across all classes in the training dataset.

TABLE 2. Class distribution in the training dataset.

Class ID	Class Name	Number of Instances
0	pistol	1510
1	smartphone	715
2	knife	2277
3	wallet	601
4	bill	477
5	card	279

The class distribution is intentionally retained because the dataset contains both weapon and visually similar non-weapon classes. Pistols and knives constitute the target

classes, whereas smartphones, wallets, banknotes, and cards act as hard-negative categories. These objects may share shape, scale, or handling characteristics with weapons in surveillance imagery, making discrimination more difficult [10], [32], [33].

The inclusion of these hard-negative categories supports a safety-oriented evaluation in which false-positive behavior is analyzed under visually ambiguous conditions. This is particularly relevant in low-resolution, partially occluded, or unfavorable-viewpoint scenarios, where lightweight detectors may generate unstable predictions. The dataset therefore provides an appropriate benchmark for evaluating the ability of ACTF to suppress erroneous system-level alarms rather than only maximizing balanced frame-level classification performance.

B. SYSTEM OVERVIEW

The proposed system performs continuous weapon detection and alarm confirmation using two concurrent camera streams acquired from spatially separated viewpoints. The architecture comprises two video sources, a shared deep learning detector applied independently to each stream, and a decision-level temporal fusion module.

Each camera frame is processed independently using a TensorRT-optimized YOLO detector. The resulting target-class detections, including class labels, confidence scores, and acquisition timestamps, are forwarded to the ACTF module, which integrates confidence, persistence, and cross-camera temporal consistency to produce the final alarm state.

The cameras are not assumed to be hardware synchronized. Instead, temporal compatibility is evaluated using frame timestamps and a predefined cross-camera validation interval. This formulation allows detections acquired at slightly different times to be associated while rejecting evidence that exceeds the permitted temporal separation.

The system is designed for computationally constrained edge platforms, where detection accuracy, processing throughput, and alarm latency must be considered jointly. ACTF operates exclusively on the detector outputs and does not introduce an additional neural-network inference stage. Consequently, its computational cost is small compared with image acquisition, preprocessing, and detector inference, although the complete application throughput remains dependent on the hardware platform and camera pipeline.

Figure 1 summarizes the complete dual-camera processing architecture. The two camera streams are acquired concurrently and processed independently by identical YOLO26s TensorRT engines. Detector-level outputs are subsequently combined by ACTF using confidence separation, temporal persistence, class compatibility, and cross-camera timestamp validation.

C. DETECTION MODEL

Object detection is performed using a YOLO-based detection model trained to identify multiple object classes. The model processes each input frame independently and outputs a set of detections defined as

$$\mathcal{D} = \{(b_i, c_i, s_i)\}_{i=1}^{N}, \tag{1}$$

where b_i represents the bounding-box coordinates, c_i denotes the predicted class label, s_i is the associated confidence score, and N is the number of candidate detections in the frame.

The trained detectors were optimized using TensorRT to reduce inference latency and support continuous embedded operation on NVIDIA Jetson platforms. This optimization modifies the execution representation of the network but does not alter the learned detector parameters or the ACTF decision rules.

Only detections corresponding to the target weapon classes, pistol and knife, are forwarded to the fusion stage. When multiple target detections are present in the same frame, the detection with the highest confidence score is retained for each camera stream. This selection reduces the amount of information processed by ACTF and provides one confidence–class pair per stream and processing cycle.

Because the cameras observe the scene from different viewpoints, their predictions may differ due to occlusion, object scale, illumination, and perspective. ACTF is therefore used as a system-level validation layer that filters unstable evidence without modifying the underlying detection architecture.

D. MODEL SELECTION AND ARCHITECTURAL CONSIDERATIONS

To evaluate lightweight object detectors under embedded computing constraints, this study considers Ultralytics YOLO26 and YOLOv10 architectures in their nano and small configurations. Specifically, YOLO26n, YOLO26s, YOLOv10n, and YOLOv10s were trained and evaluated under the same dataset partitions, image resolution, training protocol, and evaluation conditions.

Ultralytics YOLO26 is a unified real-time vision model family designed for efficient end-to-end deployment on accelerated and resource-constrained platforms [34]. Its detection architecture uses a dual-head formulation composed of a one-to-one inference head and a one-to-many training head. The default one-to-one head directly produces a maximum of 300 detections per image in the format of bounding-box coordinates, confidence score, and class index, without requiring a separate Non-Maximum Suppression stage. This output configuration was preserved during ONNX export and TensorRT deployment.

YOLO26 also removes Distribution Focal Loss from bounding-box regression and incorporates MuSGD, Progressive Loss, and Small-Target-Aware Label Assignment. Progressive Loss gradually shifts training supervision toward the one-to-one inference branch, whereas Small-Target-Aware Label Assignment improves positive-label coverage for small objects, which is relevant when weapons occupy a limited image area.

Although YOLOv10 also supports end-to-end NMS-free detection, it is treated as a separate comparison architecture. YOLOv10 relies on consistent dual assignments for NMS-free inference, whereas YOLO26 combines its dual-head design with DFL-free regression and updated optimization and assignment mechanisms. Therefore, the evaluated models represent distinct detector families rather than alternative names for the same architecture.

The nano and small variants were selected to examine the trade-off among detection performance, model capacity, memory demand, and inference latency. Evaluating these models under identical conditions also allows the contribution of ACTF to be analyzed separately from detector-specific improvements.

E. ADAPTIVE MULTI-CAMERA TEMPORAL FUSION

To improve alarm reliability while introducing minimal additional computation, this work proposes the Adaptive Multi-Camera Temporal Fusion strategy. ACTF operates at the decision level and combines confidence-aware evidence separation, temporal persistence, and cross-camera confirmation.

The term adaptive refers to the state-dependent treatment of the available evidence. Weak, strong, persistent, and cross-camera detections produce different state transitions and alarm responses. ACTF does not perform online optimization of its thresholds during inference; the operating parameters are defined before evaluation.

1) CONFIDENCE SEPARATION

Each target-class detection is categorized according to two confidence thresholds:

$$W_j(t) = \mathbb{I}\left[s_j(t) \geq T_w\right], \tag{2}$$
$$S_j(t) = \mathbb{I}\left[s_j(t) \geq T_s\right], \tag{3}$$

where $j \in \{0, 1\}$ identifies the camera stream, T_w is the weak evidence threshold, T_s is the strong evidence threshold, and $T_s > T_w$.

Weak evidence contributes to the activation of a preliminary alarm state, whereas strong evidence is required for cross-camera confirmation. Separating these confidence levels prevents an isolated low-confidence prediction from immediately generating a final alarm.

2) TEMPORAL PERSISTENCE

To suppress isolated weak detections, ACTF maintains a pre-alert persistence counter. Let $w_i(t)$ denote the weak-evidence indicator for camera i at processing cycle t, defined as $w_i(t) = \mathbb{I}[s_i(t) \geq T_w]$. The dual-camera weak-evidence condition is therefore

$$W(t) = w_0(t) \vee w_1(t). \tag{4}$$

The pre-alert persistence counter $r_p(t)$ is updated recursively as

$$r_p(t) = \begin{cases} r_p(t-1) + 1, & \text{if } W(t) = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The pre-alert state is activated when the weak-evidence condition remains active for at least N_p consecutive processing cycles:

$$P(t) = \mathbb{I}\left[r_p(t) \geq N_p\right]. \tag{6}$$

Thus, N_p represents the number of consecutive camera-pair processing cycles containing weak target-class evidence. The counter is reset when neither camera produces a confidence value above T_w, preventing isolated or interrupted weak detections from activating a persistent pre-alert.

3) MULTI-CAMERA CONFIRMATION

Strong evidence is defined by $q_i(t) = \mathbb{I}[s_i(t) \geq T_s]$, where $T_s > T_w$. Because the two camera streams are acquired concurrently but are not hardware synchronized, ACTF does not require strong detections to occur within exactly the same processing cycle. Instead, a current strong detection from one camera is compared with the most recent strong detection retained from the complementary camera.

Let $\tau_i(t)$ and $\hat{c}_i(t)$ denote, respectively, the timestamp and predicted class of the most recent strong detection produced by camera i. Cross-camera compatibility is defined as

$$C(t) = \mathbb{I}\Big[\big(q_0(t) = 1 \wedge t - \tau_1(t) \leq \Delta t \wedge c_0(t) = \hat{c}_1(t)\big) \vee \big(q_1(t) = 1 \wedge t - \tau_0(t) \leq \Delta t \wedge c_1(t) = \hat{c}_0(t)\big)\Big]. \tag{7}$$

Therefore, $C(t) = 1$ when a strong detection from either camera is supported by a class-consistent strong detection from the other camera whose timestamp is no more than Δt seconds old. This formulation accommodates small acquisition and processing delays between the two camera streams while preventing temporally distant detections from being associated.

F. DECISION STRATEGY

ACTF maintains a confirmation accumulator $r_c(t)$ to avoid activating the final alarm from a single cross-camera coincidence. The accumulator is updated according to

$$r_c(t) = \begin{cases} r_c(t-1) + 1, & \text{if } P(t) = 1 \wedge C(t) = 1, \\ \max\left(0, r_c(t-1) - 1\right), & \text{otherwise.} \end{cases} \tag{8}$$

Unlike a strict consecutive-frame rule, the confirmation accumulator decreases gradually when the cross-camera condition is not satisfied. This behavior tolerates brief interruptions in detector evidence while ensuring that sustained inconsistencies eventually remove the confirmed state. The accumulator is reset when neither camera provides weak evidence.

The final decision is defined as

$$Y(t) = \begin{cases} 1, & \text{if } r_c(t) \geq N_c, \\ 0.5, & \text{if } P(t) = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where N_c is the confirmation-accumulator threshold required to activate the final alarm. The three output values represent a confirmed alarm, a pre-alert, and the absence of valid evidence, respectively.

IV. EXPERIMENTAL SETUP

The proposed system was evaluated under controlled dual-camera surveillance conditions using embedded edge-computing platforms. The scenario-based detection evaluation of ACTF was conducted on an NVIDIA Jetson Nano, whereas an additional computational benchmark was performed on an NVIDIA Jetson Xavier NX to analyze the effect of hardware capacity on complete-pipeline throughput. Both implementations used two camera streams acquired at a resolution of 640 × 480 pixels and a TensorRT-optimized YOLO26s detector with an input resolution of 640 × 640 pixels. The scenario-level detection results correspond to the original Jetson Nano evaluation, while the Xavier NX experiment was used exclusively for computational performance analysis.

The evaluation considered alarm reliability, false-positive suppression, frame-level detection continuity, computational throughput, and alarm-confirmation latency. For clarity, the Jetson Nano results are described as low-rate online embedded operation. The term *reactive real-time embedded operation* is used in this study for a complete dual-camera pipeline sustaining approximately 10 or more camera-pair processing cycles per second. This operational definition is distinguished from conventional full-video-rate processing at approximately 25–30 FPS.

Figure 1 illustrates the complete processing architecture, including concurrent camera acquisition, timestamped frame preprocessing, independent TensorRT inference, confidence separation, temporal persistence, cross-camera confirmation, and final alarm generation.

A. SCENARIO-BASED QUANTITATIVE PERFORMANCE ANALYSIS

The evaluation was conducted using controlled video sequences representing common surveillance challenges. The scenario recordings were independent of the Sohas training, validation, and test subsets and were acquired exclusively for the system-level evaluation of ACTF. Table 3 summarizes the evaluated conditions and their corresponding metrics.

1) HARDWARE AND IMPLEMENTATION DETAILS

The original scenario-based implementation was deployed on an NVIDIA Jetson Nano equipped with a quad-core ARM Cortex-A57 CPU and 4 GB of RAM. The additional performance benchmark was conducted on an NVIDIA Jetson Xavier NX Developer Kit equipped with a six-core NVIDIA Carmel ARMv8.2 64-bit CPU, a 384-core NVIDIA Volta GPU with 48 Tensor Cores, and 8 GB of LPDDR4x RAM. The Xavier platform ran Jetson Linux R35.6.4, CUDA 11.4, and TensorRT 8.5.2.2 and was configured using the `20W 6CORE` power mode during all performance measurements.

TABLE 3. Evaluated surveillance scenarios and performance metrics.

Category	Description
Scenarios	Clear detection; partial occlusion; distractor-induced false positives; challenging illumination
Detection metrics	Precision; recall; F1-score; missed frames; false positives; specificity
Computational metrics	Camera-pair processing rate; per-camera processed FPS; aggregate image rate; detector-only throughput; alarm latency

Two USB cameras were connected concurrently to each platform. Both cameras acquired images at 640 × 480 pixels. The streams were not assumed to be hardware synchronized; instead, acquisition timestamps were recorded and temporal compatibility was evaluated by the ACTF cross-camera validation mechanism. For the Xavier experiments, camera acquisition was executed in independent threads to prevent sequential frame-reading delays.

The trained YOLO26s checkpoint was first exported to ONNX using a static batch size of one and an input tensor of 1 × 3 × 640 × 640. A TensorRT FP16 engine was then generated locally on each target platform because serialized TensorRT engines are dependent on the software and hardware environment. The engine retained the native end-to-end output tensor of $1 \times 300 \times 6$, where each prediction contains four bounding-box coordinates, one confidence score, and one class index.

Before performance measurement, the TensorRT engine was executed for 20 warm-up iterations to exclude initialization overhead from the reported measurements. Detector-only throughput was evaluated separately from the complete application using TensorRT benchmarking, whereas complete-pipeline measurements included camera acquisition, image resizing and normalization, two sequential detector executions, ACTF processing, visualization, CSV logging, and video encoding.

On the Jetson Xavier NX, the isolated TensorRT FP16 engine achieved 55.42 inferences per second with a mean inference latency of 18.59 ms. The complete dual-camera implementation achieved an average of 10.2 camera-pair processing cycles per second, corresponding to 10.2 processed frames per second for each camera stream and 20.4 camera images per second in aggregate.

The acquisition resolution of 640×480 pixels was selected because it provides a practical balance between visual detail, USB bandwidth, memory use, and embedded processing cost. Under the controlled geometry employed in this study, the

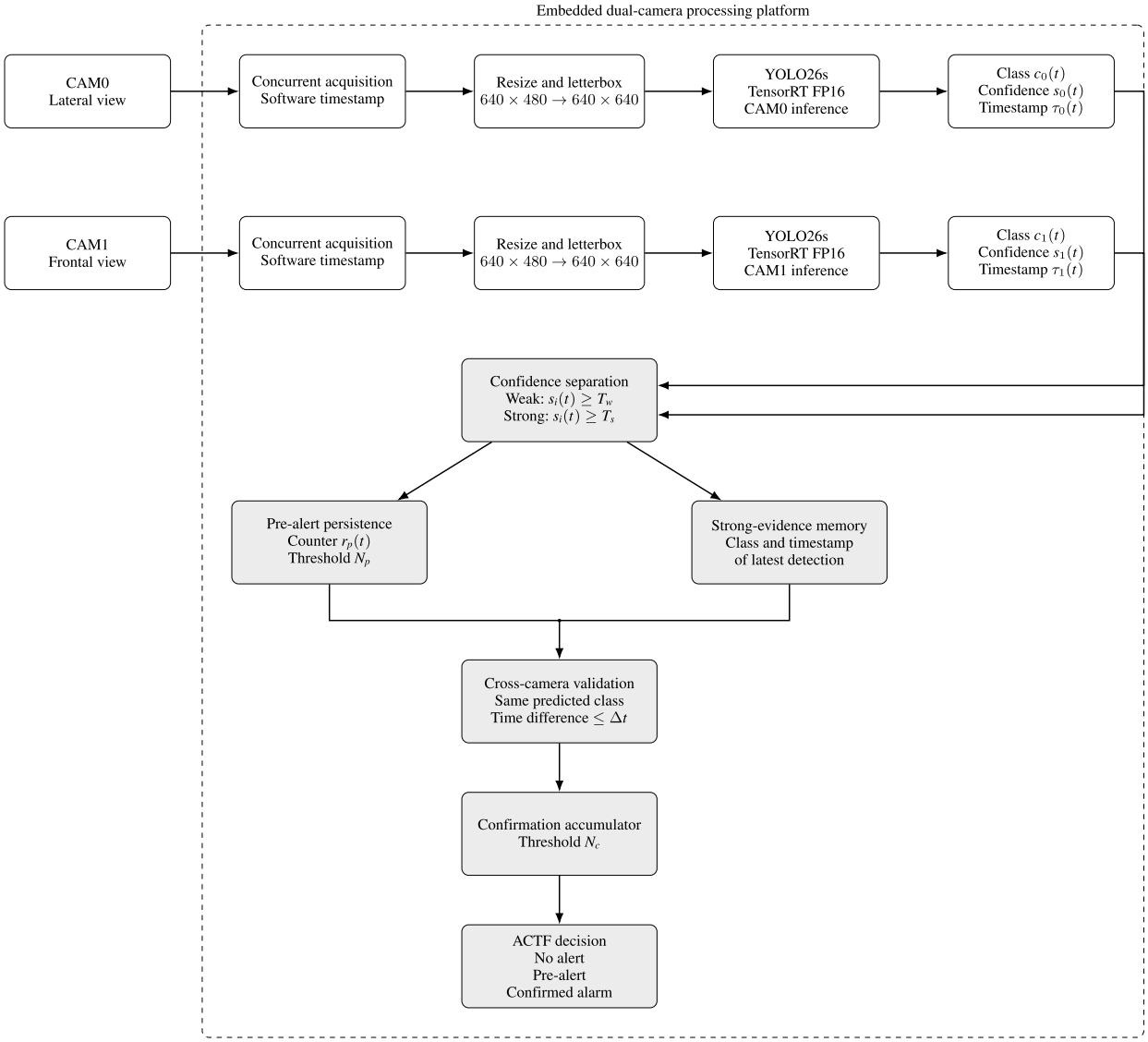


FIGURE 1. System architecture of the proposed dual-camera ACTF pipeline. CAM0 and CAM1 are acquired concurrently and assigned software timestamps. Each stream is independently preprocessed and processed by the same TensorRT-optimized YOLO26s detector. Detector outputs are passed to ACTF, which applies confidence separation, pre-alert persistence, strong-evidence memory, cross-camera validation, and a confirmation accumulator to generate the final alarm state.

subject was located between 1 and 2 m from the cameras, allowing the weapon to remain visually distinguishable in at least one viewpoint. Each acquired image was resized and letterboxed to the detector input resolution of 640 × 640 pixels.

The selected acquisition resolution was considered adequate for the controlled distances and viewpoints evaluated in this work; however, it should not be interpreted as sufficient for all surveillance conditions. Long-range operation, very small weapon instances, and highly crowded environments may require higher-resolution sensors or small-object enhancement techniques and are outside the scope of the present evaluation.

Table 4 summarizes the geometric configuration of the dual-camera surveillance setup used during experimental evaluation. The cameras were intentionally positioned to provide complementary frontal and lateral viewpoints in order to maximize viewpoint diversity for multi-camera validation while preserving realistic surveillance geometry.

TABLE 4. Geometric configuration of the experimental dual-camera surveillance setup.

Parameter	CAM0 (Lateral)	CAM1 (Frontal)
Mounting Height (m)	2.20	1.80
Distance to Subject (m)	2.00	1.00
Viewpoint Orientation	Lateral	Frontal
Acquisition Resolution	640 × 480	640 × 480
Primary Role in Fusion	Side Validation	Front Confirmation

The experimental setup was calibrated considering an average subject height of approximately 1.80 m to ensure realistic

framing and weapon visibility across both viewpoints during the evaluated surveillance scenarios.

2) ACTF CONFIGURATION

The reference ACTF configuration was defined as

$$\left(T_w, T_s, N_p, N_c, \Delta t\right) = (0.30, 0.45, 2, 2, 0.6\ \text{s})\,. \tag{10}$$

The weak-confidence threshold $T_w = 0.30$ determines whether a detector response contributes preliminary evidence to the pre-alert state, whereas the strong-confidence threshold $T_s = 0.45$, with $T_s > T_w$, identifies detections eligible for cross-camera confirmation. This separation prevents low-confidence observations from directly activating the final alarm while allowing them to initiate a pre-alert when they persist over time.

The parameter $N_p = 2$ specifies the number of consecutive camera-pair processing cycles containing weak evidence in at least one camera that are required to activate the pre-alert state. The associated pre-alert counter increases while weak evidence remains available and is reset when neither camera produces a detection with confidence greater than or equal to T_w.

The parameter $N_c = 2$ denotes the confirmation-accumulator threshold required for final alarm activation. Unlike N_p, this parameter does not represent a strict number of consecutive cross-camera confirmations. The confirmation accumulator increases when the pre-alert state and the cross-camera compatibility condition are simultaneously satisfied and decreases by one, with a lower bound of zero, when this condition is interrupted. The accumulator is reset when both camera streams fall below the weak-confidence threshold.

The cross-camera validation interval $\Delta t = 0.6$ s defines the maximum permitted temporal separation between a current strong detection from one camera and the most recent class-consistent strong detection from the complementary camera. The interval is evaluated using timestamps rather than frame indices because the cameras are acquired concurrently but are not hardware synchronized and the processing rate may vary between cycles.

At the average Jetson Nano processing rate of 2.4 camera-pair cycles per second, the 0.6-s temporal interval spans approximately 1–2 processed camera-pair cycles. At the average Jetson Xavier NX rate of 10.2 camera-pair cycles per second, the same interval spans approximately six cycles. These conversions are provided only to facilitate interpretation; ACTF evaluates temporal compatibility directly in seconds and does not convert Δt into a fixed number of frames.

The same detector configuration, confidence thresholds, temporal parameters, and decision logic were retained in the Xavier NX computational benchmark. Therefore, differences in the reported processing rates are attributed to the embedded hardware and complete-pipeline implementation rather than to changes in the ACTF operating configuration.

3) PARAMETER SENSITIVITY PROTOCOL

A one-factor-at-a-time sensitivity analysis was conducted to evaluate the influence of the ACTF confidence and temporal parameters. The reference configuration was $(T_w, T_s, N_p, N_c, \Delta t) = (0.30, 0.45, 2, 2, 0.6\ \text{s})$. During each experiment, one parameter was varied while the remaining parameters were maintained at their reference values.

The evaluated values were $T_w \in \{0.20, 0.30, 0.40\}$, $T_s \in \{0.40, 0.45, 0.50, 0.60\}$, $N_p \in \{1, 2, 3\}$, $N_c \in \{1, 2, 3\}$, and $\Delta t \in \{0.2, 0.4, 0.6, 0.8\}$ s. The analysis was performed offline using the recorded detector confidence scores, predicted classes, timestamps, and ground-truth conditions from the clear-detection, challenging-illumination, and distractor-based false-positive sequences. The detector was not retrained or executed again; therefore, the observed variations are attributable exclusively to the ACTF decision parameters.

Frame-level recall was evaluated for the positive sequences, whereas the false-positive count and specificity were evaluated using the distractor-based negative sequence. Macro recall was calculated as the arithmetic mean of the recall obtained in the clear-detection and challenging-illumination sequences. The reference configuration was retained from the original implementation and was not retrospectively optimized using the sensitivity results.

B. EVALUATION PROTOCOL

The proposed system was evaluated using four detector and fusion configurations: CAM0, CAM1, logical OR fusion, and the proposed ACTF strategy. These configurations were tested under four controlled surveillance scenarios, and the evaluation metrics were selected according to the ground-truth composition of each sequence.

Figure 2 summarizes the evaluated configurations, controlled scenarios, and corresponding metric assignment. Positive scenarios were evaluated using recall and missed detections, whereas the distractor-based negative scenario was evaluated using false positives and specificity. Computational throughput and temporal response were analyzed across the complete evaluation.

1) EVALUATION SCENARIOS

The system was evaluated across the following surveillance scenarios:

- **Scenario 1 (Clear Detection):** The weapon is clearly visible in both camera views under normal illumination conditions.
- **Scenario 2 (Partial Occlusion):** The weapon is visible in one camera while partially occluded or not detected in the second camera.
- **Scenario 3 (False-Positive Conditions):** Visually ambiguous non-weapon objects, including mobile phones and wallets, are introduced while no weapon is present.

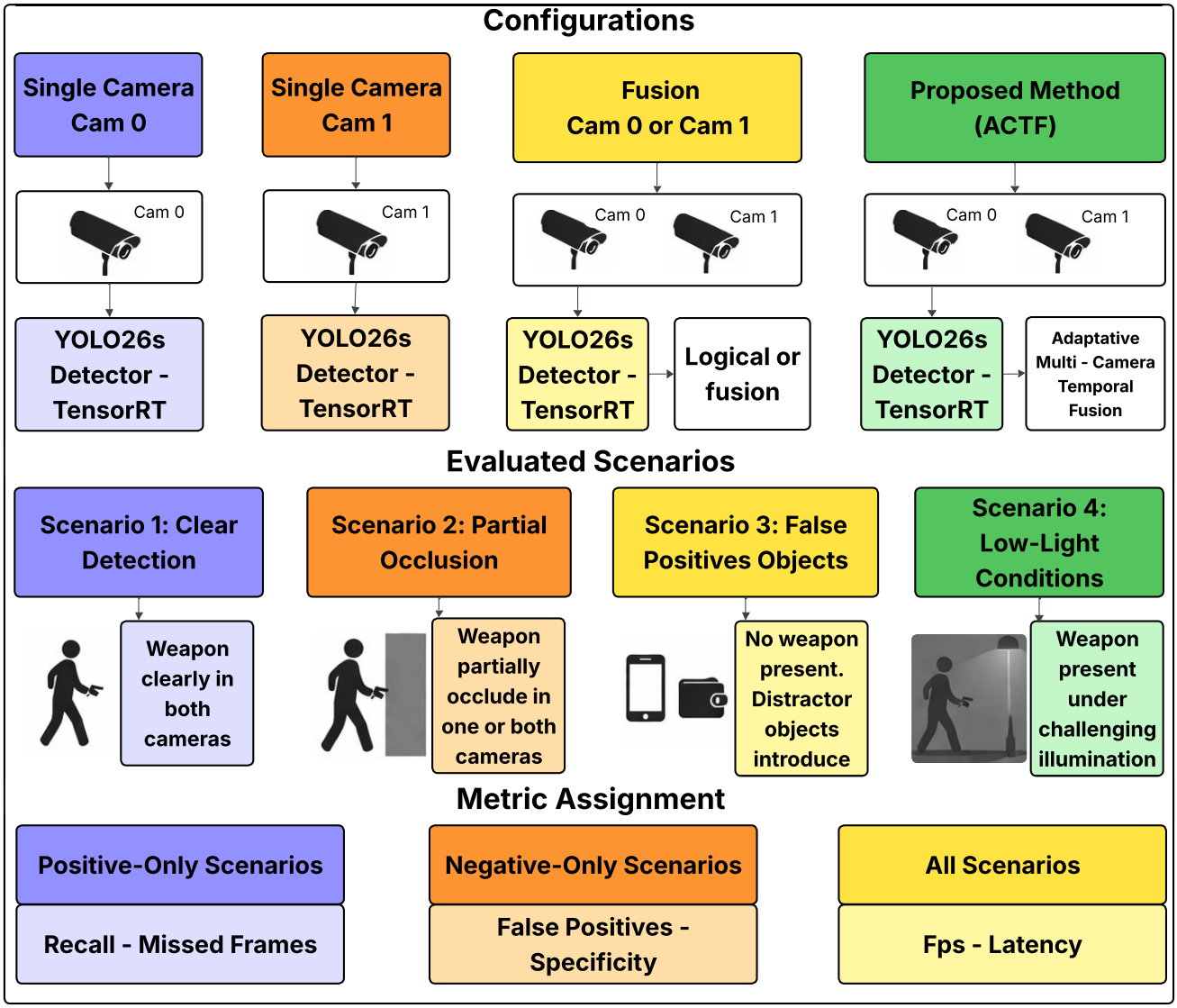


FIGURE 2. Experimental evaluation protocol. The four evaluated configurations include independent CAM0 and CAM1 detection, logical OR fusion, and the proposed ACTF strategy. The configurations were analyzed under clear detection, partial occlusion, distractor-based false-positive conditions, and challenging illumination. Metrics were assigned according to the ground-truth composition of each scenario.

- **Scenario 4 (Challenging Illumination):** A weapon is present under reduced or uneven illumination to evaluate detection stability.

The camera streams were acquired concurrently rather than through hardware-level synchronization. Each frame was associated with its acquisition timestamp, and cross-camera evidence was considered temporally compatible only when the difference between the relevant detection timestamps did not exceed Δt.

The ground-truth condition was defined before each recording according to the controlled scenario protocol and was subsequently verified by one researcher through manual inspection of the recorded sequences. All processed frame pairs in the clear-detection, partial-occlusion, and challenging-illumination scenarios were labeled as weapon-present because a weapon remained in the monitored scene throughout each recording. All processed frame pairs in the distractor-based false-positive scenario were labeled as weapon-absent because no weapon was introduced during that recording.

Table 5 reports the duration, number of processed frame pairs, and ground-truth composition of each recording. These values refer to the live scenario sequences and not to the Sohas image test set.

Each scenario was configured with a nominal acquisition duration of 60 s. The number of processed frame pairs differs among sequences because the embedded pipeline operated at a variable processing rate depending on camera acquisition, inference, and scene conditions. Each frame pair represents one complete processing cycle containing one observation from CAM0 and one observation from CAM1. Representative views of the dual-camera experimental configuration are shown in Figure 3.

TABLE 5. Composition of the scenario-based dual-camera evaluation sequences.

Scenario	Nominal duration (s)	Processed frame pairs	Weapon-present pairs	Weapon-absent pairs
Clear detection	60	148	148	0
Partial occlusion	60	192	192	0
False-positive conditions	60	177	0	177
Challenging illumination	60	187	187	0

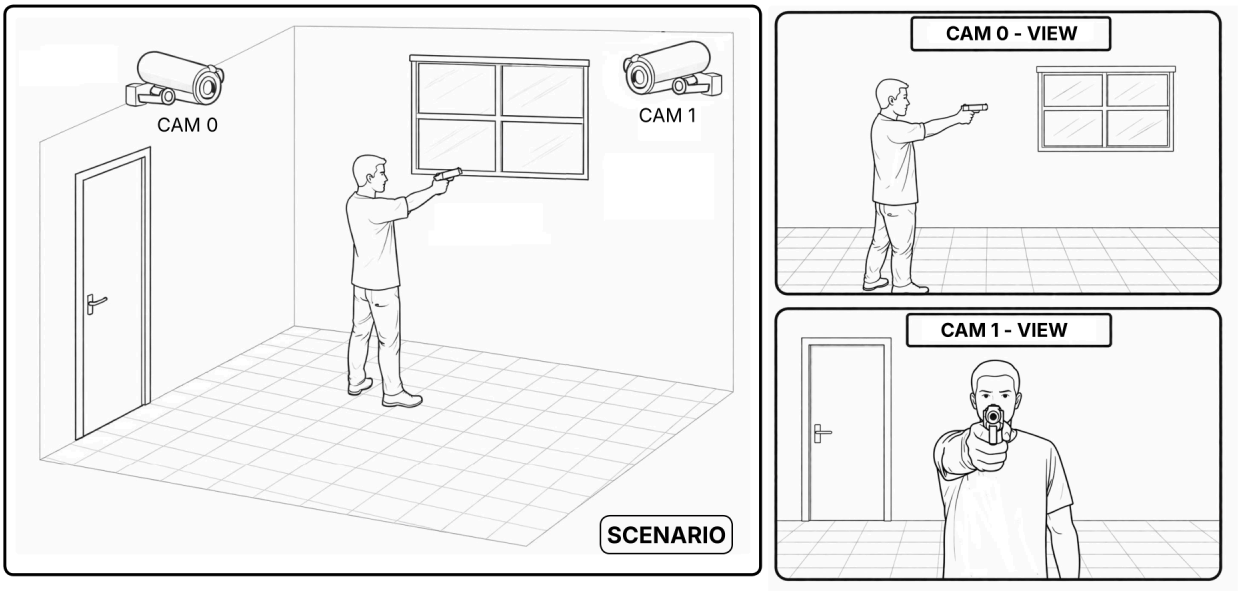


FIGURE 3. Representative lateral and frontal views of the dual-camera experimental configuration. CAM0 provides a lateral viewpoint, whereas CAM1 provides a complementary frontal viewpoint of the monitored area.

2) EVALUATION METRICS

The system was evaluated at the frame-pair and event levels. Metrics were selected according to the ground-truth composition of each scenario. Recall and missed frames were used in positive scenarios, whereas false positives and specificity were used in the weapon-absent distractor scenario.

- **Precision:** Proportion of positive weapon predictions corresponding to weapon-present ground-truth observations.
- **Recall:** Proportion of weapon-present observations producing a positive detection or confirmed alarm.
- **F1-score:** Harmonic mean of precision and recall when both positive and negative observations are available.
- **False Positives:** Number of weapon detections or alarm activations occurring during weapon-absent observations.
- **Specificity:** Proportion of weapon-absent observations correctly classified as negative.
- **Camera-Pair Processing Rate:** Number of complete dual-camera processing cycles executed per second. Each cycle includes one frame from CAM0 and one frame from CAM1.
- **Per-Camera Processed Frame Rate:** Effective number of processed frames per second for each individual camera stream. In the dual-camera implementation, this value is equal to the camera-pair processing rate.
- **Aggregate Camera-Image Rate:** Total number of camera images processed across both streams per second, calculated as twice the camera-pair processing rate.
- **Detector-Only Throughput:** Number of TensorRT inferences executed per second without camera acquisition, visualization, logging, video encoding, or ACTF processing.
- **End-to-End Alarm Latency:** Elapsed time between the manually annotated first visible appearance of a weapon and the first activation of the ACTF confirmed-alarm state.

For an annotated event e, the end-to-end alarm latency is defined as

$$L_e = t_{\text{alarm},e} - t_{\text{onset},e}, \tag{11}$$

where $t_{\text{onset},e}$ is the ground-truth weapon appearance time and $t_{\text{alarm},e}$ is the time at which ACTF first enters the confirmed-alarm state. Mean alarm latency is calculated over the set of successfully confirmed events rather than over individual positive frames. Events that are not confirmed are reported as missed events and are not assigned an artificial latency value.

These definitions distinguish detector-only inference speed from complete application throughput and prevent the aggregate two-camera image rate from being interpreted as the frame rate of each individual stream.

V. RESULTS

A. EXPERIMENTAL EVALUATION OVERVIEW

The proposed system was evaluated under controlled dual-camera surveillance conditions using an embedded edge computing platform. The objective of the evaluation is to analyze both detector-level performance and system-level reliability, with particular emphasis on false-positive behavior, temporal consistency, and real-time operational constraints.

To ensure a comprehensive analysis, the evaluation is structured in two complementary stages. First, the performance of multiple lightweight YOLO-based detectors is assessed to characterize their detection capabilities and inherent limitations. This stage provides insight into precision–recall trade-offs and the susceptibility of different architectures to false-positive activations.

Second, the proposed ACTF strategy is evaluated at the system level and compared against baseline configurations. This enables direct quantification of the contribution introduced by temporal persistence and multi-camera confirmation mechanisms in improving alarm reliability.

The following subsections present the experimental results, beginning with detector-level analysis, followed by system-level evaluation under controlled surveillance scenarios.

B. DETECTOR PERFORMANCE COMPARISON

To analyze the impact of detector architecture on system-level performance, multiple lightweight YOLO-based models were evaluated under identical training conditions, including YOLO26 and YOLOv10 variants in both nano and small configurations.

Table 6 summarizes the detection performance of all evaluated models.

TABLE 6. Performance comparison of lightweight YOLO models under identical training conditions.

Model	Precision	Recall	mAP@50	mAP@50-95
YOLO26n	0.861	0.899	0.937	0.813
YOLO26s	0.913	0.920	0.954	0.847
YOLOv10n	0.895	0.855	0.930	0.785
YOLOv10s	0.919	0.869	0.953	0.825

The results indicate that YOLO26 variants achieve higher recall values compared to YOLOv10 models, suggesting greater sensitivity to potential weapon instances. However, this increased sensitivity may also lead to a higher susceptibility to false-positive detections.

In contrast, YOLOv10 models exhibit higher precision, indicating more conservative detection behavior with fewer false alarms but potentially reduced sensitivity under challenging conditions.

Additionally, small variants consistently outperform nano variants across all metrics, confirming that increased model capacity improves detection stability and overall accuracy.

These observations highlight the inherent trade-off between detection sensitivity and reliability in lightweight detectors, reinforcing the need for system-level validation mechanisms such as the proposed ACTF framework.

C. CONFUSION MATRIX ANALYSIS

To further analyze detector-level performance, a confusion matrix was generated using the best-performing model (YOLO26s), which serves as the baseline for system-level evaluation.

Figure 4 illustrates the classification performance across all object categories. The results reveal the presence of misclassifications between weapon classes and visually similar non-weapon objects, particularly smartphones and wallets.

These misclassifications highlight the inherent difficulty of distinguishing weapons from everyday objects under conditions representative of common surveillance challenges, especially in scenarios involving low resolution, partial occlusion, or challenging viewpoints.

This behavior explains the occurrence of false-positive detections in single-camera and naive fusion approaches, and reinforces the need for additional validation mechanisms beyond detector-level improvements. In this context, the proposed ACTF framework plays a critical role in filtering ambiguous detections and improving overall alarm reliability.

D. SCENARIO-BASED PERFORMANCE EVALUATION

To evaluate system behavior under controlled surveillance conditions, the proposed ACTF strategy was tested across four representative scenarios: (i) clear detection, (ii) partial occlusion, (iii) false-positive conditions, and (iv) low-light environments. These scenarios were designed to provide conditions representative of common surveillance challenges, including viewpoint variation, visual ambiguity, and illumination changes.

The performance of ACTF was compared against three baseline configurations: (i) single-camera detection using CAM0, (ii) single-camera detection using CAM1, and (iii) dual-camera logical OR fusion without temporal validation. These baselines allow isolating the contribution of temporal persistence and multi-camera confirmation.

Frame-level temporal analysis was performed over complete video sequences for each scenario. Since ground-truth

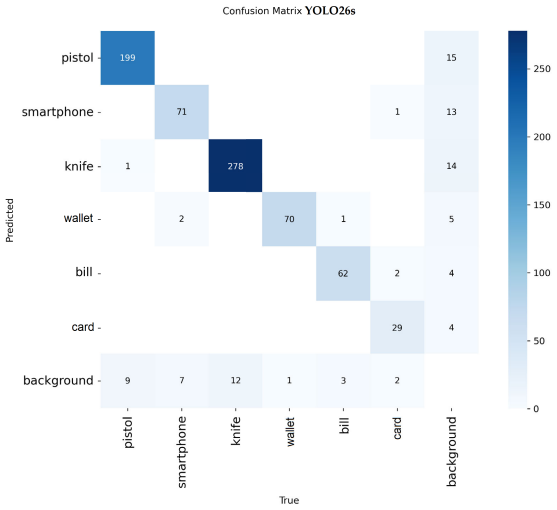


FIGURE 4. Confusion matrix of the YOLO26s model on the test dataset. Misclassifications between weapon and visually similar non-weapon objects contribute to false-positive detections in surveillance scenarios.

conditions vary across scenarios, the evaluation metrics were selected accordingly. In positive-only scenarios, recall and missed frames were used to quantify detection continuity over time. In the false-positive scenario, where no weapon is present, false positives and specificity were used to assess false alarm suppression capability.

Table 7 summarizes the frame-level performance across all evaluated configurations.

As shown in Table 7, naive OR fusion consistently achieved the highest frame-level recall across all positive scenarios, indicating that it maximizes temporal detection sensitivity by triggering detections whenever at least one camera produces a positive response.

However, this increased sensitivity comes at the cost of reliability. In the false-positive scenario, OR fusion produced 12 false-positive frame-level activations, whereas ACTF produced no confirmed alarms during the negative sequence. This behavior highlights a fundamental limitation of naive multi-camera fusion strategies, where maximizing sensitivity directly increases the likelihood of false alarm generation.

In contrast, the proposed ACTF strategy completely suppressed false alarms in the distractor scenario while maintaining competitive recall under clear visibility conditions. Although ACTF introduces a reduction in recall in more challenging scenarios, this reflects a deliberate design choice that prioritizes alarm reliability over raw temporal sensitivity.

By enforcing temporally consistent multi-camera confirmation, ACTF effectively filters transient and ambiguous detections, transforming a high-sensitivity but error-prone detection system into a more stable and reliable surveillance solution. In safety-critical applications, this trade-off must be managed carefully because false-alarm suppression should not come at the cost of systematically masking valid single-view detections under degraded viewpoints.

E. FALSE ALARM SUPPRESSION ANALYSIS

The most significant advantage of the proposed ACTF strategy was observed in the distractor-based false-positive scenario, where visually ambiguous objects such as smartphones and wallets were intentionally introduced to challenge the detection system.

While CAM1 and OR fusion each generated 12 false positive detections, the proposed ACTF strategy produced zero false alarms throughout the entire evaluation sequence. This result confirms that the temporal persistence and multi-camera confirmation mechanisms effectively suppress transient and ambiguous detections that would otherwise propagate through naive fusion rules.

Figure 5 illustrates the temporal alarm activation patterns observed during the false-positive scenario, highlighting the suppression of spurious activations by the proposed ACTF strategy.

F. TEMPORAL BEHAVIOR OF ACTF

To further characterize the temporal behavior of the proposed confirmation mechanism, additional ACTF-specific response metrics were analyzed, as summarized in Table 8.

The reported delay represents the elapsed time between the first activation of the pre-alert state and the first confirmed ACTF alarm within each scenario. It should not be interpreted as end-to-end latency measured from the physical appearance of the weapon. The previously reported Mean Delay metric was removed because its temporal reference and averaging population were not sufficiently defined.

ACTF state transitions indicate the number of changes between non-confirmed and confirmed alarm states during each sequence. The larger number of transitions observed under partial occlusion and low illumination reflects the instability of the detector evidence under degraded observation conditions.

The measured confirmation delays indicate that ACTF maintains sub-second response behavior in favorable operating conditions while introducing longer confirmation times when evidence becomes inconsistent or intermittent. This behavior is expected, as the proposed method intentionally delays alarm generation until sufficient temporal and multi-camera evidence has been accumulated.

Moreover, the reduced number of ACTF state transitions compared to raw detector outputs indicates improved temporal stability and lower flickering behavior in the final alarm stream.

G. ACTF PARAMETER SENSITIVITY ANALYSIS

A one-factor-at-a-time sensitivity analysis was conducted using the recorded confidence scores, class predictions, timestamps, and ground-truth conditions from the clear-detection, challenging-illumination, and distractor-based false-positive sequences. During each experiment, one ACTF parameter was varied while the remaining parameters were

TABLE 7. Scenario-level frame-based performance of the evaluated configurations.

Scenario	Metric	Single CAM0	Single CAM1	OR Fusion	ACTF
Clear Detection	Recall	0.905	0.811	0.919	0.905
	Missed Frames	14	28	12	14
Partial Occlusion	Recall	0.464	0.406	0.688	0.375
	Missed Frames	103	114	60	120
False Positive Conditions	False Positives	0	12	12	0
	Specificity	1.000	0.932	0.932	1.000
Low-Light Conditions	Recall	0.401	0.567	0.754	0.439
	Missed Frames	112	81	46	105

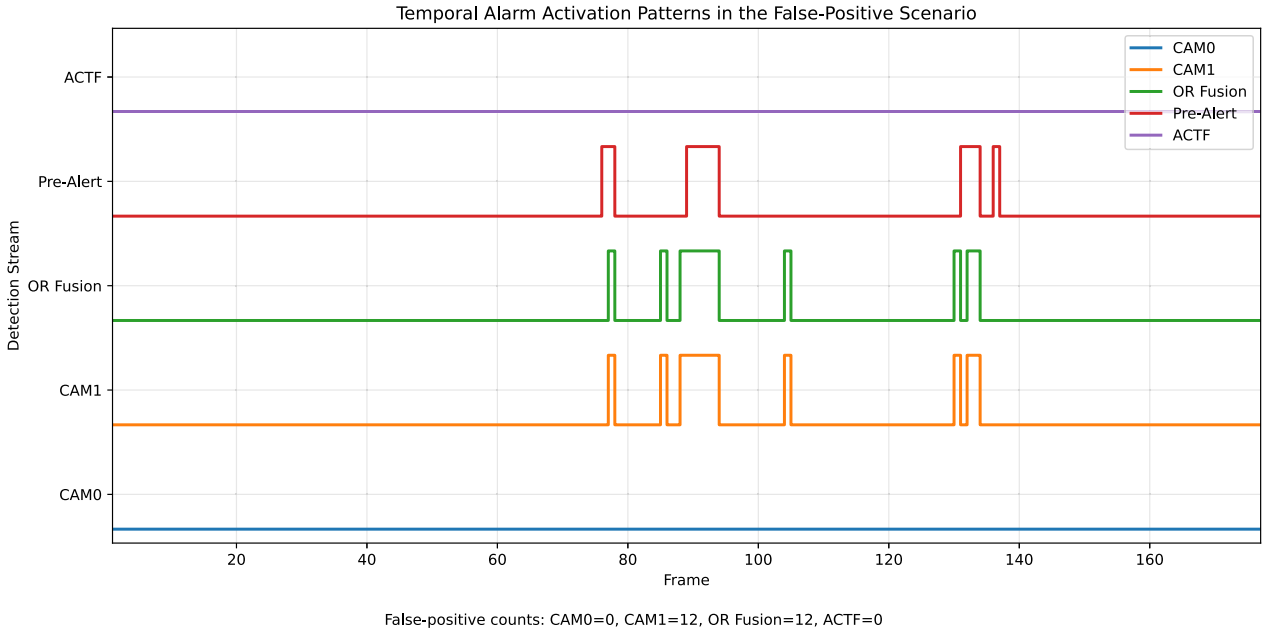


FIGURE 5. Temporal alarm activation patterns in the false-positive scenario. While CAM1 and OR fusion produced spurious activations in response to distractor objects, the proposed ACTF strategy suppressed all false alarms by enforcing temporal persistence and multi-camera confirmation before issuing a final alarm.

TABLE 8. Temporal response characteristics of ACTF.

Scenario	Pre-alert-to-confirmation delay (s)	ACTF state transitions
Clear Detection	0.307	1
Partial Occlusion	1.187	13
Low Light	0.878	14

fixed at the reference configuration $(T_w, T_s, N_p, N_c, \Delta t) = (0.30, 0.45, 2, 2, 0.6\ \text{s})$. The original detector outputs were reused; therefore, the observed variations are attributable exclusively to the ACTF decision parameters.

Table 9 shows that the clear-detection sequence was comparatively insensitive to the evaluated parameter ranges, maintaining a recall close to 0.90 in all configurations. The challenging-illumination sequence exhibited greater sensitivity. Reducing T_w from 0.30 to 0.20 increased its recall from 0.439 to 0.519, whereas increasing T_w to 0.40 reduced recall to 0.385. Similarly, increasing T_s from 0.45 to 0.60 reduced the challenging-illumination recall from 0.439 to 0.369.

The temporal parameters produced the expected reliability–sensitivity trade-off. Reducing the confirmation threshold to $N_c = 1$ increased macro recall from 0.672 to 0.707, whereas $N_c = 3$ reduced it to 0.642. The cross-camera validation interval had the strongest temporal effect: reducing Δt to 0.2 s decreased macro recall to 0.584, whereas increasing it to 0.8 s raised macro recall to 0.680.

Across all evaluated configurations, ACTF generated zero confirmed false-positive frames in the distractor sequence and maintained a specificity of 1.000. Less restrictive settings, particularly $T_w = 0.20$ and $N_c = 1$, improved recall without producing confirmed false positives in the available negative sequence. Nevertheless, the reference configuration was retained because it was defined before the reported evaluation and the sensitivity analysis was not

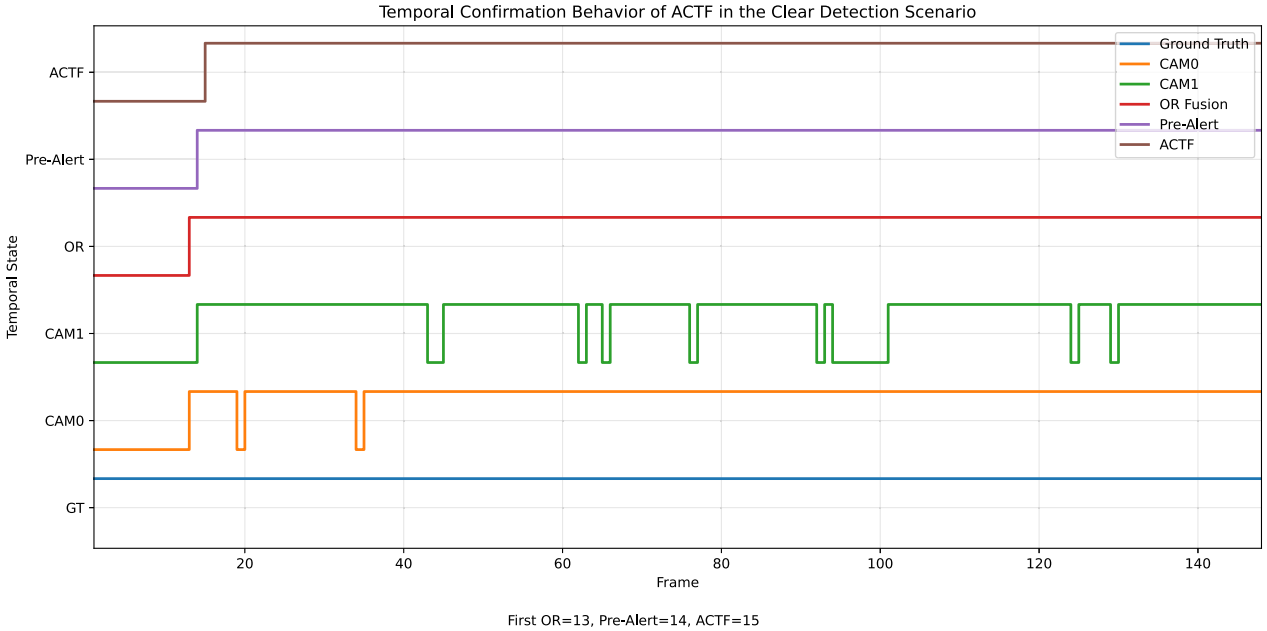


FIGURE 6. Temporal confirmation behavior of the proposed ACTF strategy in the clear detection scenario. The figure illustrates how raw detections from both cameras are aggregated through OR fusion, pre-alert activation, and final ACTF confirmation. The temporal confirmation logic introduces a short validation delay before issuing the final alarm, resulting in a more stable and temporally consistent detection stream.

used to retrospectively optimize the method. These results characterize the behavior of the ACTF parameters rather than establish a universally optimal configuration.

H. EMBEDDED THROUGHPUT AND OPERATIONAL RESPONSIVENESS

The original NVIDIA Jetson Nano implementation achieved a complete dual-camera processing rate ranging from 1.75 to 2.82 camera-pair cycles per second, with an average of approximately 2.4 pairs per second. Because each processing cycle includes one frame from each camera, this corresponds to approximately 2.4 processed FPS per camera stream and 4.8 camera images per second in aggregate. This result is characterized as low-rate online embedded operation rather than conventional video-rate real-time processing.

To assess the effect of a more capable embedded platform, an additional benchmark was conducted on the NVIDIA Jetson Xavier NX using the same YOLO26s TensorRT FP16 detector, camera resolutions, and ACTF configuration. The complete dual-camera pipeline achieved an average processing rate of 10.2 camera-pair cycles per second, equivalent to 10.2 processed FPS per camera stream and 20.4 camera images per second in aggregate. This represents a 4.25-fold increase relative to the average Jetson Nano camera-pair rate.

The isolated TensorRT engine achieved 55.42 inferences per second, with a mean inference latency of 18.59 ms and a mean GPU compute time of 18.03 ms. The difference between detector-only throughput and complete-pipeline throughput is associated with dual-camera acquisition, image preprocessing, sequential inference of both camera streams, visualization, metric logging, and video encoding. ACTF operates on scalar class, confidence, and temporal information and therefore introduces negligible computational overhead relative to detector inference and multimedia processing.

Under the operational terminology adopted in this work, the Xavier NX implementation supports reactive real-time embedded processing because it sustains more than 10 complete dual-camera processing cycles per second. This classification remains explicitly distinct from conventional 25–30 FPS full-video-rate surveillance processing.

The Jetson Xavier NX achieved a complete dual-camera processing rate of 10.2 camera-pair cycles per second, representing a 4.25-fold improvement over the average rate obtained with the Jetson Nano. Since every processing cycle contains one frame from each camera, the measured pair rates are equivalent to 2.4 and 10.2 processed FPS per camera for the Jetson Nano and Xavier NX, respectively. The corresponding aggregate image-processing rates were 4.8 and 20.4 images per second.

I. SUMMARY OF KEY FINDINGS

The experimental results highlight the following key findings:

- Naive OR fusion maximizes temporal recall but propagates false detections from individual cameras.
- The proposed ACTF strategy completely suppresses false alarms in distractor-based scenarios.

TABLE 9. One-factor-at-a-time sensitivity analysis of the ACTF parameters.

Parameter	Value	Clear recall	Low-light recall	Macro recall
T_w	0.20	0.905	0.519	0.712
Reference	**--**	**0.905**	**0.439**	**0.672**
T_w	0.40	0.905	0.385	0.645
T_s	0.40	0.905	0.449	0.677
T_s	0.50	0.905	0.406	0.656
T_s	0.60	0.905	0.369	0.637
N_p	1	0.905	0.449	0.677
N_p	3	0.899	0.422	0.661
N_c	1	0.912	0.503	0.707
N_c	3	0.899	0.385	0.642
Δt	0.20	0.905	0.262	0.584
Δt	0.40	0.905	0.369	0.637
Δt	0.80	0.905	0.455	0.680

Note: False-positive (FP) and specificity columns are omitted because all evaluated sensitivity configurations produced identical results in the available distractor sequence (FP = 0; specificity = 1.000).

TABLE 10. Embedded computational performance comparison.

Metric	Jetson Nano	Jetson Xavier NX
Camera-pair rate (pairs/s)	2.40	10.20
Processed rate per camera (FPS)	2.40	10.20
Aggregate image rate (images/s)	4.80	20.40
Relative pair-rate improvement	1.00×	4.25×

- ACTF improves temporal stability and reduces alarm flickering through persistence-based confirmation.
- The additional Xavier NX benchmark achieved 10.2 camera-pair cycles per second, representing a 4.25-fold throughput improvement over the Jetson Nano implementation, while ACTF continued to introduce negligible decision-level computational overhead.
- ACTF is best interpreted as a robust alarm confirmation mechanism rather than a sensitivity-maximizing fusion rule.

The experimental results demonstrate that the operational performance of an embedded weapon-surveillance system cannot be fully characterized using conventional detector-level metrics alone. Although YOLO-based architectures may achieve high precision and recall under standard image-based evaluation protocols, the scenario-based experiments showed that individual detections remain susceptible to visually ambiguous objects, viewpoint degradation, partial occlusion, and reduced illumination.

The confusion-matrix analysis supports this observation by showing misclassifications between weapon classes and visually similar non-weapon objects, including smartphones and wallets. In practical surveillance applications, these errors are more critical than isolated classification mistakes because they may directly trigger unnecessary alarm events. Similar limitations have been reported in intelligent surveillance systems, where visual ambiguity, occlusion, and environmental variability degrade detection reliability [10], [19].

From a system-level perspective, logical OR fusion increased detection sensitivity because an alarm was generated whenever either camera produced a positive response. However, the same rule propagated incorrect detections generated by individual viewpoints. Although OR fusion achieved the highest recall in several positive scenarios, it also generated 12 false-positive responses in the distractor-based sequence. This result confirms that increasing the number of camera observations does not necessarily improve alarm reliability when the fusion mechanism does not evaluate temporal persistence or cross-camera compatibility.

In contrast, ACTF combines weak-evidence persistence, strong-evidence validation, class compatibility, and timestamp-based cross-camera confirmation. Transient detections that are not supported by persistent and temporally compatible evidence remain at the pre-alert level and do not activate the final alarm. This behavior explains why ACTF produced no confirmed false-positive responses in the evaluated distractor sequence, whereas logical OR fusion propagated the errors generated by individual cameras.

The false-alarm suppression obtained by ACTF is consistent with the general findings of established temporal post-processing strategies. Kong et al. demonstrated that averaging frame-level probabilities over a temporal interval can suppress isolated false detections in fire surveillance [27]. Their results also showed that increasing the smoothing interval introduces a sensitivity–latency trade-off, because stronger temporal smoothing reduces short confidence fluctuations but delays the response to a true event. ACTF follows the same general objective of rejecting transient responses; however, it does not replace the detector confidence with a moving average. Instead, weak and strong evidence remain separated and are used for different alarm states.

Recursive state estimation provides another established mechanism for reducing temporal noise. Wang et al. applied an online Kalman filter to image-quality features for camera-anomaly detection [28]. Their filtered implementation reduced false positives from seven to two and increased precision from 88.89% to 96.55%, while maintaining a recall of 98.25%. These results demonstrate

that a recursively estimated state can suppress unstable measurements without introducing substantial computational overhead. Nevertheless, Kalman filtering assumes that the monitored feature follows an appropriate state-transition and measurement model. A semantically incorrect weapon detection may remain temporally smooth and therefore would not necessarily be rejected by state estimation alone.

The use of two or more confidence levels in ACTF is also supported by established hysteresis-based surveillance methods. Yeh et al. proposed a three-pronged framework combining color and texture background modeling, spatial compensation through hysteresis thresholding, and temporal compensation using motion history [29]. Their method generates a predominant foreground representation using a restrictive threshold and multiple supplementary representations using progressively less restrictive thresholds. Supplementary evidence is incorporated only when it is spatially connected to reliable foreground information, preventing unrestricted acceptance of noisy low-threshold responses.

The quantitative evaluation reported by Yeh et al. showed that the combined hysteresis and compensation strategy achieved competitive or improved precision, recall, and F-measure over several established background-subtraction methods. For an object moving toward the camera, their method achieved a recall of 0.970, a precision of 0.901, and an F-measure of 0.933. In the Cubicle sequence, it achieved a precision of 0.990 and an F-measure of 0.874. These results demonstrate that hysteresis is not an improvised thresholding rule, but an established strategy for balancing restrictive and permissive evidence in real-time surveillance systems.

ACTF adopts this general hysteresis principle at a different semantic level. The weak threshold T_w allows uncertain detections to contribute to a pre-alert, whereas the strong threshold T_s identifies detections that are eligible for final cross-camera validation. Unlike the method of Yeh et al., ACTF does not reconstruct foreground regions or propagate neighboring pixels. It operates on predicted classes, confidence values, persistence counters, and timestamps. Therefore, ACTF can be interpreted as a decision-level and multi-camera extension of the general hysteresis principle rather than as a conventional pixel-level hysteresis algorithm.

Gao et al. also demonstrated the relevance of hysteresis and spatio-temporal post-processing for false-detection reduction [30]. Their spatial stage combines multi-neighborhood hysteresis thresholding with morphological opening and closing, whereas their temporal stage evaluates the consistency of detected objects between adjacent registered frames using intersection-over-union and centroid-distance criteria. The complete scheme achieved an average reduction of 83.8% in false positives and improved the average F-score of the evaluated vehicle-detection algorithms.

The method of Gao et al. and ACTF share the use of temporal consistency and differentiated evidence, but their operating requirements differ. The aerial-vehicle method requires spatial registration and geometric correspondence between objects in adjacent frames. ACTF does not require bounding-box overlap, centroid tracking, or geometric calibration between camera views. Instead, it evaluates whether class-consistent strong evidence from one camera is supported by recent evidence from the complementary camera within Δt. This decision-level formulation reduces computational complexity but does not provide object-trajectory estimation.

Zaidi et al. used another temporal strategy based on sliding-window voting for construction-safety monitoring [31]. Their system triggers an alarm when at least 70% of the frames within a 5-s window are classified as unsafe. Across five evaluated videos, the temporal module increased average monitoring accuracy from 97.97% to 100% by preventing isolated missed or false detections from directly generating alarms. This confirms that frame-level predictions should not always be interpreted as independent alarm decisions.

However, a fixed 5-s voting window is most appropriate when the monitored condition remains stable over several seconds. A weapon event may be shorter or may require a more immediate response. ACTF therefore uses recursive state updates and a shorter timestamp-based compatibility interval of $\Delta t = 0.6$ s instead of waiting for a fixed multi-second window to be completed. This increases responsiveness, although the stricter requirement for evidence from both viewpoints can reduce sensitivity when one camera is degraded.

Taken together, these five studies demonstrate that temporal smoothing, recursive filtering, hysteresis, motion-history compensation, spatial consistency, and sliding-window voting are established methods for reducing false alarms in surveillance systems. The principal distinction of ACTF is therefore not the use of temporal information by itself, but the source and interpretation of that information. The reviewed methods primarily regularize observations within a single stream or between adjacent frames. ACTF additionally requires class-consistent strong evidence from two complementary camera viewpoints before confirming the final alarm.

The parameter-sensitivity analysis further demonstrated that ACTF provides a controllable trade-off between detection sensitivity and alarm reliability. The clear-detection sequence remained relatively stable throughout the evaluated parameter ranges, with recall values close to 0.90. In contrast, the challenging-illumination sequence was more sensitive to the confidence thresholds. Reducing T_w from 0.30 to 0.20 increased its recall from 0.439 to 0.519, whereas increasing T_w to 0.40 reduced recall to 0.385. Similarly, increasing T_s to 0.60 reduced the challenging-illumination recall to 0.369.

The temporal parameters produced a similar reliability–sensitivity relationship. Reducing the confirmation accumulator threshold from $N_c = 2$ to $N_c = 1$ increased macro recall from 0.672 to 0.707, whereas increasing it to $N_c = 3$ reduced macro recall to 0.642. The cross-camera validation interval produced the strongest temporal effect. Reducing Δt from

0.6 s to 0.2 s decreased macro recall to 0.584 because fewer detections satisfied the temporal compatibility requirement. Increasing Δt to 0.8 s increased macro recall to 0.680 by tolerating a larger inter-camera delay.

All evaluated sensitivity configurations produced zero confirmed false-positive frames and maintained a specificity of 1.000 in the available distractor sequence. Nevertheless, this result should not be interpreted as evidence of universally false-alarm-free operation. The negative sequence was controlled and limited in duration, and all tested configurations were evaluated using the same available set of distractors. Longer recordings and a wider variety of non-weapon objects may produce different false-positive behavior.

The reference configuration was retained even though some less restrictive settings achieved greater recall without generating false positives in the available negative sequence. The reference parameters were established before the sensitivity analysis and were not retrospectively optimized using the evaluation recordings. The analysis therefore characterizes how ACTF behaves as its operating point changes rather than establishing a universally optimal parameter set.

The improvement in false-alarm suppression was accompanied by a relevant reduction in recall under degraded viewpoints. In the partial-occlusion scenario, ACTF achieved a recall of 0.375, compared with 0.688 for logical OR fusion. Under challenging illumination, ACTF achieved a recall of 0.439, whereas logical OR fusion achieved 0.754. These differences constitute an important operational limitation rather than only a numerical trade-off. A genuine threat detected consistently by one camera may not reach the final alarm state when the complementary viewpoint is affected by occlusion, poor illumination, or an unfavorable viewing angle.

For safety-critical deployment, we recommend a two-level alarm configuration rather than using strict ACTF confirmation as the sole alarm criterion. Persistent single-camera weapon evidence that reaches the pre-alert state should generate a secondary warning requiring operator verification, whereas persistent, class-consistent, and temporally compatible evidence from both viewpoints should activate the confirmed ACTF alarm. This deployment configuration preserves the false-alarm suppression benefits of strict cross-camera confirmation while reducing the risk of masking valid single-view detections when the complementary camera is affected by partial occlusion, poor illumination, or an unfavorable viewpoint. Therefore, ACTF-only confirmation is not recommended as the sole deployment mode in environments where degraded viewpoints are expected.

Temporal filtering and ACTF should consequently be viewed as complementary rather than mutually exclusive strategies. Moving-average or Kalman-based smoothing could be applied to detector confidences to reduce short measurement fluctuations before ACTF processing. Tracking or motion-history compensation could help recover brief detector interruptions. ACTF would then remain responsible for final alarm validation using independent multi-camera evidence. Evaluating such hybrid combinations represents an important direction for future work.

A key finding of this work is that alarm reliability can be improved without modifying or retraining the underlying detector. ACTF operates as a lightweight decision-level layer and does not introduce an additional neural-network inference stage. Consequently, the method can be incorporated into existing detector pipelines without increasing model size or substantially changing inference cost.

From a practical perspective, reducing false alarms is particularly important in continuously operated surveillance systems. Frequent incorrect alerts may increase operator workload, reduce trust in the system, and cause valid alarms to receive less attention. However, the results also demonstrate that false-alarm suppression should not be optimized independently from detection sensitivity. The operational configuration must balance the consequences of a false alarm against the consequences of failing to confirm a genuine threat.

The computational experiments show that complete-pipeline throughput is strongly dependent on the embedded platform. The Jetson Nano implementation achieved an average of approximately 2.4 camera-pair cycles per second and is therefore characterized as continuous low-rate online operation. With the same YOLO26s detector, TensorRT FP16 precision, camera resolutions, and ACTF parameters, the Jetson Xavier NX achieved an average of 10.2 camera-pair cycles per second. This corresponds to 10.2 processed frames per second for each camera stream and represents a 4.25-fold improvement over the Jetson Nano implementation.

The difference between isolated TensorRT throughput and complete dual-camera throughput indicates that detector inference is not the only computational cost. Camera acquisition, image preprocessing, per-stream detector execution, visualization, video encoding, and metric logging also contribute to the complete processing cycle. Since ACTF processes only predicted classes, confidence values, timestamps, and small temporal accumulators, its computational overhead remains negligible relative to detector inference and multimedia operations.

Several limitations must be acknowledged. First, the scenario-based evaluation was conducted under controlled conditions and does not fully represent the duration, scene diversity, crowd density, camera motion, or distractor variability of operational surveillance environments.

In addition, each experimental condition was evaluated using a single continuous recording with a nominal duration of 60 s, and the reported scenario-level measurements were not averaged over repeated independent trials. Frame pairs within each recording are temporally correlated and therefore should not be interpreted as statistically independent samples. No confidence intervals were calculated for these measurements; consequently, the reported scenario-level values should be interpreted as descriptive results for the evaluated recordings rather than population-level performance estimates.

Second, the available negative sequence cannot establish that zero false alarms would be maintained during long deployments. Third, strict cross-camera confirmation reduces recall when one viewpoint is degraded. Fourth, the temporal comparison with moving average, Kalman filtering, hysteresis, and sliding-window voting is currently methodological and based on the published literature rather than on a common experimental dataset. Finally, the cameras were acquired concurrently but were not hardware synchronized, and temporal compatibility was managed using software timestamps and the interval Δt.

Overall, the results indicate that combining complementary camera viewpoints with temporal decision validation can substantially improve alarm reliability. Nevertheless, ACTF should be interpreted as a configurable reliability layer rather than a universally superior temporal filter or sensitivity-maximizing fusion rule. Future work should evaluate longer operational recordings, additional distractor categories, hardware-synchronized acquisition, camera-quality-aware confirmation, the recommended two-level warning configuration under more diverse operational conditions, and direct experimental comparisons with moving-average, Kalman-filter, hysteresis, sliding-window, and tracking-based temporal strategies using the same recorded sequences.

VI. CONCLUSION

This work presented a dual-camera surveillance system for continuous weapon detection and reactive alarm confirmation, integrating a lightweight YOLO-based detector with an Adaptive Multi-Camera Temporal Fusion strategy. The proposed approach addresses a key challenge in practical surveillance systems by reducing false-positive alarms under visually ambiguous conditions.

Experimental results demonstrated that, despite the strong performance of modern object detectors in terms of precision and recall, they remain susceptible to false detections caused by visually similar non-weapon objects. This limitation becomes more critical in real-time deployments, where erroneous detections directly affect alarm reliability.

The proposed ACTF framework effectively mitigated this issue by enforcing temporal consistency and multi-camera validation prior to alarm activation. Unlike naive multi-camera fusion approaches, which prioritize detection sensitivity, ACTF significantly improved alarm reliability by filtering transient and ambiguous detections. In the evaluated scenarios, the method successfully eliminated false alarms while maintaining competitive performance under favorable visibility conditions.

An important contribution of this work is the demonstration that system-level reliability can be improved without modifying the underlying detector architecture. By operating as a lightweight decision-level module, ACTF preserves computational efficiency across embedded platforms. The Jetson Nano implementation supported continuous low-rate online processing, whereas the additional Jetson Xavier NX benchmark achieved an average of 10.2 camera-pair cycles per second, supporting reactive real-time embedded operation.

These findings highlight the importance of shifting the focus from purely detector-centric evaluation toward integrated system-level design. In safety-critical applications, reliable alarm generation requires balancing false-alarm suppression with preservation of valid detections when one camera operates under a degraded viewpoint.

Future work will explore adaptive fusion strategies that dynamically adjust confirmation thresholds based on scene context and detection confidence. Extending the proposed approach to multi-camera networks with more than two viewpoints, as well as incorporating contextual and semantic information, represents a promising direction for improving robustness and scalability.

Overall, this study demonstrates that combining multi-camera configurations with temporal validation provides an effective and practical solution for enhancing the reliability of real-time intelligent surveillance systems.

Nevertheless, the strict cross-camera confirmation requirement reduced recall under partial occlusion and challenging illumination because valid evidence from one camera was not always confirmed by the complementary viewpoint. This limitation must be considered in safety-critical deployments.

Accordingly, a two-level deployment configuration is recommended in which persistent single-view evidence that reaches the pre-alert state generates a secondary warning requiring operator verification, while strict cross-camera ACTF confirmation is reserved for the highest-confidence confirmed-alarm state. This configuration reduces the risk of masking genuine single-view detections when the complementary camera is affected by occlusion, poor illumination, or an unfavorable viewpoint.

The sensitivity analysis showed that the confidence thresholds, confirmation accumulator, and cross-camera validation interval control the trade-off between recall and alarm reliability, with degraded illumination exhibiting the greatest parameter sensitivity.

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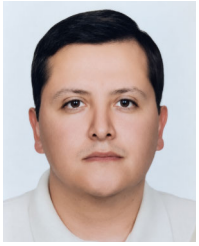
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